GOOGLE WORKSPACE TRAINING



Google Sheets Advanced

Two-day course review draft

Build and check practical workplace outputs with fictional project data. These notes and prompts are a written review draft awaiting full live lab validation.

Delivered by Asraf · Microsoft Certified Trainer

Google Sheets Advanced

Two-day course review draft · Review draft, version 0.4

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Resources: Use the practice files supplied with this review package.

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Screenshots show the product at the time of writing. Google updates its apps often, so labels, availability and layouts in your account may differ. This review draft still requires full lab validation.

About the author

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He holds a Bachelor of Information Technology (Multimedia) from the University of Malaya, along with Microsoft certifications in Power BI (PL-300) and AI for business (AB-730). Outside the classroom he writes his own automation tools in Python and Node.js, mostly to save himself from doing the same task twice.

He runs his training practice solo, which means he has personally felt every inefficiency this book tries to fix.

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INTRODUCTION

Before you begin

This review book accompanies Google Sheets Advanced. Work through its chapters in order and keep the original practice inputs unchanged. The same notes and prompts also appear in the local course website.

Current Status

All chapters in this book are written drafts. Most have not yet been executed end to end in the application. The masterclass's reference brief chapter has a tested core path with a recorded recovery prompt; that does not validate the other exercises. Treat VERIFY notes and draft notices as production work still to complete.

What You Need

Use a desktop browser and the Google account approved for the class. Confirm app and AI access beforehand. Edition, administrator controls and language can affect availability.

Download the supplied fictional practice pack. No real client or personnel record is needed. Dataset week fields are relative practice weeks, not an agreed pilot start date.

How to Use the Exercises

Follow the numbered steps, attempt the output and then open the expected-result box. AI wording can vary; source facts and approval conditions must remain correct.

Values and prompts follow the notes. Replace source placeholders before submitting. Inspect proposed edits before applying them. Drafting does not authorise sending, inviting people or publishing.

Course Timing

This course is planned for 2 days, with 390 teaching minutes per day, excluding breaks and lunch. Estimates include practice, checks and discussion and need a classroom pilot.

Each course supplies independent starting inputs. Use a stated recovery file if an earlier output is missing.

Expanded Casework

Revision 0.4 adds casework, independent challenges and evidence gates to every chapter. The original short guided path alone is insufficient. Timings are targets, not measured learner results. Read each case input and keep baseline and scenario files separate.

01

CHAPTER 1

Audit the Source Register

In this chapter you will check types, blanks, duplicates and keys before analysis.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 45 minutes: 10 explanation/demonstration, 15 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 35 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: An audited source, preserved snapshot and empty Analysis workspace.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Inspect dataset grain and keys.
- Check types and baseline controls.
- Preserve the source before analysis.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Download ProcurementRegister.xlsx. It contains 24 fictional order lines and a six-item rate table, separate from the pilot budget.

1.1 Prepare the Workbook

1. Upload the workbook and save a native Google Sheets copy named `GW - Sheets Advanced - Analysis`.
2. Open Orders. Confirm headers in A1:I1 and records PO001 to PO024 in rows 2:25.
3. Open Rates and confirm six unique item codes in A2:A7.

What should you see?

Each Orders row is one fictional order line, not a project subtotal or a real purchase commitment.

1.2 Audit Key Fields

1. Check OrderID values for blanks and duplicates. Use a temporary COUNTIF diagnostic if needed.
2. Inspect Quantity and UnitCost: they must be numeric. Week is a practice index, not a date.
3. Compare the status vocabulary Delivered, Pending and Cancelled. Keep cancelled records visible for analysis.

What should you see?

The baseline has 24 unique order IDs and the expected categories.

1.3 Protect the Analysis Trail

1. Duplicate Orders as `Source Snapshot` before adding calculations.
2. Create a new tab called `Analysis` for summaries.
3. Record the source range and meaning of an order line in a note outside the working table.

What should you see?

The original supplied values can be compared with later analysis without guessing what changed.

1.4 Quarantine Bad Rows and Write a Data Contract

1. Create **Audit Intake** and preserve the six anomaly records from the copy-paste card separately from Orders. Add Issue, Evidence and Disposition columns.
2. Define one row's grain, OrderID uniqueness, valid ItemCode mapping, allowed status vocabulary and numeric Quantity/UnitCost requirements in a **Data Contract** section.
3. Use COUNTIF to inspect duplicate IDs and an exact lookup or manual comparison for unknown item keys. Keep a repeated record distinct from a conflicting duplicate.
4. Use ISNUMBER or equivalent inspected checks to distinguish numeric 4 from text "four". Document whether zero is valid for the stated training rule.
5. Compare cancelled and pending records with the contract. A cancelled row is not automatically corrupt or removable from an all-orders population.
6. Create an exception register with a proposed next action for each anomaly: clarify, correct with evidence, retain as valid or quarantine pending evidence.
7. Recheck the original 24 unique IDs and six Rates keys after the audit. Keep intake anomalies out of baseline controls.

What should you see?

Data contract and six-row anomaly audit with preserved raw values. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently decide whether each anomaly should enter the analysis, remain pending or be retained as a valid case. Justify every decision against the contract.
2. Write a release condition for the source: what must be true before conditional calculations and lookups can be trusted?
3. Have a reviewer challenge one disposition and resolve it using source evidence or an explicit unresolved label.

Evidence to Hand Over

- Data contract and six-row anomaly audit with preserved raw values.
- Baseline uniqueness/type/key checks and a justified quarantine policy.
- Independent release conditions and a recorded challenged decision.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A cancelled row is removed as dirty data	Separate data validity from the inclusion rule of the analysis question.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

An analysis begins with data meaning, types and source coverage.

Check yourself: What does one row represent, and which field identifies it?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
GW - Sheets Advanced - Analysis
```

```
Source Snapshot
```

```
Analysis
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Separate intake anomalies: duplicate PO001 with changed quantity; missing OrderID; ItemCode IT99; Quantity “four”; valid zero quantity; valid Cancelled status. Do not inject these into Orders. Other fields can be copied from a source row only in Audit Intake and labelled as practice.
```

02

CHAPTER 2

Build Conditional Calculations

In this chapter you will revise mixed references and apply IF, SUMIF, COUNTIF, SUMIFS and COUNTIFS with test cases.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 75 minutes: 15 explanation/demonstration, 32 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 60 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Checked line amounts, a project/status summary grid and tested logic bands.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Build row calculations.
- Apply conditional totals with mixed references.
- Test a nested rule at its boundaries.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders A1:I25 and the Analysis tab from Chapter 1.

2.1 Calculate Line Amounts

1. In Orders J1 type `Amount`; in J2 enter `=E2*F2` and fill through J25.
2. In K1 type `ActiveAmount`; in K2 enter `=IF(G2="Cancelled",0,J2)` and fill through K25.
3. Compare J2, J13 and J25 with quantity times unit cost. Cancelled rows remain in the dataset but contribute zero to ActiveAmount.

What should you see?

The all-orders Amount sum is RM 3,158. ActiveAmount answers a different question from all-orders Amount.

2.2 Create a Summary Grid

1. On Analysis enter Project in A3 and Delivered, Pending, Cancelled in B3:D3. Put PJ-A, PJ-B and PJ-C in A4:A6.
2. In B4 enter `=SUMIFS(Orders!$J$2:$J$25,Orders!$B$2:$B$25,$A4,Orders!$G$2:$G$25,B$3)`.
3. Fill across through D4, then down through row 6. Inspect a formula in the final row and column.

What should you see?

Project criteria move down while status headers move across. The grid sums back to 3,158 across all statuses.

2.3 Test Logic Boundaries

1. In a spare practice area label the bands Low below 100, Medium from 100 to below 200, and High from 200 upward.
2. With a test amount in F2, enter `=IF(F2<100,"Low",IF(F2<200,"Medium","High"))` in G2.
3. Test 99, 100, 199 and 200 in F2 and record the classifications. Keep these hypothetical bands separate from source policy.

What should you see?

99 is Low; 100 and 199 are Medium; 200 is High. The bands do not overlap.

2.4 Reconcile Three Populations and Test Conditional Boundaries

1. Create an Analysis control block for all-orders Amount, ActiveAmount excluding Cancelled, and Delivered-only Amount. Write each population definition before calculating.
2. Build COUNTIFS as well as SUMIFS project/status grids. Reconcile the count grid to 24 and the amount grid to the all-orders source total.
3. For two grid intersections list qualifying OrderIDs and manually multiply their quantity and unit cost. Compare this independent subset with the formulas.
4. Create a separate **Logic Tests** table with amount, status, expected band and expected pending-high flag. Include 0, 99, 100, 199 and 200 and each status.
5. Apply the nested-IF band formula and the pending/amount condition to the test table. Compare expected and actual outcomes and investigate any mismatch.
6. On a grid copy remove one mixed-reference anchor and fill across/down. Locate the first incorrect criterion by inspecting the formula, then restore the anchor.
7. Record why a valid zero amount differs from an unknown item rate and why active, delivered and all-orders totals answer different questions.

What should you see?

Reconciled count/amount grids for clearly defined populations. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Build a second summary answering count and amount for Pending orders of at least 200 by project. Keep threshold logic explicit.
2. Use a filtered source list as an independent reconciliation. A zero-match baseline is a valid result; create a separate synthetic qualifying case and a non-qualifying boundary case to test the rule.
3. Change the threshold to 250 in a labelled assumption cell, repair references so the grid follows it, then restore 200 and verify restoration.

Evidence to Hand Over

- Reconciled count/amount grids for clearly defined populations.
- Expected-versus-actual boundary matrix and repaired mixed-reference defect.
- Independent threshold summary with supporting OrderIDs and a restored assumption.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
Correct totals conceal the wrong population	Reconcile counts and contributing IDs as well as the monetary total.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Visible row calculations and explicit criteria make advanced logic checkable.

Check yourself: Which references in the summary grid must stay fixed when copied across and down?

Zero-Match Check

The supplied baseline has no Pending record with Amount at least 200. Record that result explicitly. Use a separately labelled synthetic case to prove the positive branch; do not edit baseline records just to obtain a highlight.

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Amount
```

```
=E2*F2
```

```
ActiveAmount
```

```
=IF(G2="Cancelled",0,J2)
```

```
=SUMIFS(Orders!$J$2:$J$25,Orders!$B$2:$B$25,$A4,Orders!$G$2:$G$25,B$3)
```

```
=IF(F2<100,"Low",IF(F2<200,"Medium","High"))
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Logic matrix must cover below/on/above thresholds, zero quantity, Delivered, Pending and
Cancelled. Practice bands: Low <100; Medium 100 to <200; High ⩾200. Exception: Pending AND
Amount ⩾200. These are training rules, not procurement policy.
```

03

CHAPTER 3

Validate Entries and Flag Exceptions

In this chapter you will use dropdowns, custom validation and formula-based conditional formatting.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 65 minutes: 15 explanation/demonstration, 22 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 50 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Validation rules and a tested pending-order exception highlight.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Validate categorical and numeric inputs.
- Highlight exceptions using a rule.
- Test an invalid and boundary value.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use a practice copy of Orders. Restore all temporary tests before completing the chapter.

3.1 Validate New Entries

1. Select G2:G25 and add a dropdown for Delivered, Pending and Cancelled.
2. Add a validation rule to E2:E25 requiring a whole number at least 0. A zero quantity is permitted for this training dataset.
3. Inspect whether invalid values are rejected or warned about.

What should you see?

The rules reflect the stated training choices and do not silently invent a positive-only restriction.

3.2 Flag Pending High Amounts

1. Select A2:K25 and open Format, Conditional formatting.
2. Use a custom formula `=AND($G2="Pending",$J2≥200)` and a readable light fill.
3. Check the rule starts at source row 2 and applies across the intended rows.

What should you see?

Only pending rows with Amount at least 200 are highlighted; delivered or cancelled rows do not qualify.

3.3 Test the Rules

1. On one practice row, try an invalid status and a fractional quantity, then undo.
2. Test a pending row at amount 199 and 200 by temporarily changing its numeric inputs.
3. Restore the source values and inspect that the highlighting recalculates.

What should you see?

You have tested both rejection behaviour and the conditional-format threshold.

3.4 Create a Rule Test Harness and Reconcile Exceptions

1. Create **Validation Tests** with expected result and observed result before trying each input. Include whole zero, positive integer, negative, fractional and text quantities.
2. For whole nonnegative Quantity, inspect a custom rule such as `=AND(ISNUMBER(E2),E2≥0,MOD(E2,1)=0)` on a practice range. Decide and document how a blank should be handled separately.
3. Test rejection behaviour for each quantity and status case. Do not equate a coloured cell with an enforced validation rule.
4. Build an explicit exception flag column on a practice Orders copy using the same Pending/Amount logic as the conditional format.
5. Filter the flag and compare its qualifying OrderIDs with the highlighted rows. Check both the first and final source row for rule coverage.
6. On test records compare Pending at 199, 200 and 201, plus Delivered at 200 and Cancelled at 200. Explain which logical condition excludes each non-qualifying row.
7. Restore all temporary values and record the exact data-validation and conditional-format ranges used.

What should you see?

Eight-case input validation matrix and five-case exception matrix. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Append a synthetic pending record only to the test copy and diagnose whether input validation, flag formulas and conditional formatting extend to it.
2. Repair coverage and produce a test report containing at least eight input cases and five exception cases.
3. State whether blanks and pasted values were actually tested and list any behaviour still requiring confirmation in the customer account.

Evidence to Hand Over

- Eight-case input validation matrix and five-case exception matrix.
- Agreement between explicit flags, highlighted rows and filtered IDs.
- New-row coverage repair and restored source values.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The colour rule flags the wrong row	Check the apply-to start row and the relative row in the custom formula.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Data rules constrain entry; conditional formatting draws attention to defined exceptions.

Check yourself: How does a validation rule differ from a coloured warning?

Zero-Match Check

The supplied baseline has no Pending record with Amount at least 200. Record that result explicitly. Use a separately labelled synthetic case to prove the positive branch; do not edit baseline records just to obtain a highlight.

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=AND($G2="Pending",$J2⩾200)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Synthetic coverage record, test copy only: PO-TEST | PJ-A | Planning | IT01 | Quantity 5 |
UnitCost 45 | Pending | week 4 | Coverage test. Quantity candidates: 0, 2, -1, 1.5, text
“two”, blank and numeric text imported from a CSV.
```

04

CHAPTER 4

Combine and Clean Text

In this chapter you will use LEFT, RIGHT, MID and concatenation; check AI categorisation against explicit rules.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 60 minutes: 15 explanation/demonstration, 17 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Text extraction and labels with checked category mappings.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Extract parts of identifiers.
- Combine readable labels.
- Check AI categories against a visible mapping.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders and Rates. The supplied IDs follow PO plus three digits; this is a training convention.

4.1 Extract Identifier Parts

1. On a separate Text Practice tab copy OrderID and ItemCode from a few rows.
2. Use `=LEFT(A2,2)` for the prefix and `=RIGHT(A2,3)` for the numeric text.
3. Use `=MID(A2,3,3)` to extract the same three characters and compare the results. Keep leading zeros as text.

> **What should you see?**
>
> PO001 yields prefix PO and sequence text 001 rather than a newly assigned key.

4.2 Combine and Clean Text

1. Copy a space-padded team name from DirtyText.csv into a spare cell and use TRIM to clean it.
2. Use CONCATENATE or & to build a label from an order ID and project code, for example `=A2&" - "&B2` on Orders.
3. Check the underlying separate fields remain available for filtering and analysis.

What should you see?

The readable label does not replace the source columns or destroy identifier zeros.

4.3 Check Categorisation

1. Read the six-item Rates table and its Category column.
2. If Gemini is available, ask it to propose categories for the item names using only Equipment, Stationery and Print and the supplied mapping.
3. Compare every proposed category with Rates. Use the mapping as the reference; record ambiguous or unmapped inputs.

What should you see?

AI suggestions have been checked against an explicit six-item rule rather than accepted by appearance.

4.4 Build a Reversible Text Normalisation Pipeline

1. Import DirtyText.csv into **Text Audit** and preserve its raw columns. Add Normalised team, Normalised status and Decision columns.
2. Apply TRIM for spaces and use explicit case handling only where it preserves the known vocabulary. Record any non-breaking or unusual spaces that the observed function does not remove.
3. Build an identifier test table from the card. Apply LEFT, RIGHT and MID, then compare extracted values with the expected identifier pattern.
4. Flag malformed or unknown keys rather than allowing a correctly extracted substring to prove a valid mapping. Retain leading zeros as text.
5. Build a display label from OrderID and Project using CONCATENATE or & while retaining the original separate key fields.
6. Use the Rates mapping to assign the six known item categories. If AI proposes categories, compare every proposal with that mapping and retain unmapped keys as explicit exceptions.
7. Write a pipeline note showing raw → normalised → validated → category/display output. Name which transformations are reversible and which require the raw source for audit.

What should you see?

Raw/clean text audit and identifier extraction tests. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently process the mixed identifier intake into accepted keys and an exception list. Do not rely on string length alone for validity.
2. Demonstrate that IT99 is syntactically plausible but absent from the supplied mapping, and that a blank key is not a category.
3. Have a reviewer reconstruct one clean value's origin from your raw and transformation columns. Repair any missing trace.

Evidence to Hand Over

- Raw/clean text audit and identifier extraction tests.
- Separate syntax and mapping validation with explicit exceptions.
- Traceable category/display outputs and an independent mixed-intake result.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
Text extraction succeeds on a bad identifier	Validate the complete pattern and reference mapping separately from substring extraction.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Text functions reshape fields while source keys and mappings preserve meaning.

Check yourself: Why should an ID suffix such as 001 remain text?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=LEFT(A2,2)
```

```
=RIGHT(A2,3)
```

```
=MID(A2,3,3)
```

```
=A2&" - "&B2
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Identifier intake: PO001; PO024; PO1; blank; IT01; IT06; IT99; “ IT03 ”. Order IDs require PO plus three digits; mapped item keys are the six supplied Rates codes. Do not treat every extracted suffix as a valid key.
```

05

CHAPTER 5

Match Records and Handle Errors

In this chapter you will compare exact-match VLOOKUP, HLOOKUP and INDEX/MATCH; distinguish a missing key from bad input.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 80 minutes: 15 explanation/demonstration, 37 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 65 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Exact-match lookup examples and an explicit missing-item result.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Use exact-match lookups.
- Compare vertical and horizontal lookup layouts.
- Distinguish missing keys from broken formulas.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders and Rates. Rates A2:D7 has unique item codes and numeric unit costs.

5.1 Look Up a Rate

1. In a spare Orders column labelled `LookupRate`, enter `=VLOOKUP(D2,Rates!$A$2:$D$7,3,FALSE)` and fill down.
2. Compare each looked-up rate with the original F column.
3. In another labelled practice column use `=INDEX(Rates!$C$2:$C$7,MATCH(D2,Rates!$A$2:$A$7,0))` and reconcile the same result.

What should you see?

IT01 returns 45 and IT06 returns 8. Both formulas use exact matches.

5.2 Practise a Horizontal Lookup

1. On Analysis G1:L1 enter IT01 through IT06. In G2:L2 enter their unit costs 45, 12, 30, 18, 25 and 8.
2. Put IT03 in F4 and enter `=HLOOKUP(F4,$G$1:$L$2,2,FALSE)` in G4.
3. Change F4 to IT06, check 8, then restore IT03.

What should you see?

The horizontal lookup returns 30 for IT03 and uses the same explicit matching principle.

5.3 Handle an Expected Missing Key

1. In a spare test cell use item code IT99. Observe the lookup's missing-match error.
2. Wrap only the intended missing-match case, for example `=IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")`.
3. Deliberately misspell a function in another spare cell, observe the different error and repair it. Discuss how a blanket IFERROR can hide defects as well as expected missing matches.

What should you see?

An unknown item is labelled Unmapped item. A broken formula is repaired rather than treated as a zero price.

5.4 Challenge Exact Lookups With Missing and Duplicate Keys

1. Create a **Lookup Tests** table containing all six known ItemCodes, IT99, a blank and a space-padded key. Preserve raw input next to any normalised key.
2. For each known key compare exact VLOOKUP with INDEX/MATCH and the supplied Rates value. Count and investigate disagreements instead of hiding them.
3. Create **Rates Duplicate Test** as a disposable Rates copy. Append a second IT03 with a different UnitCost. Observe which result a first-match lookup returns.
4. Add a key-count diagnostic so multiple matches are flagged before accepting a looked-up amount. Record that lookup success does not prove uniqueness.
5. Reorder a copied Rates table and compare exact versus omitted-match-mode VLOOKUP. Keep approximate-match experiments clearly separated from the authoritative mapping.
6. Compare a missing-key error, a bad function name and a bad range reference. Apply IFNA to the intended missing-match case and explain how broad IFERROR could conceal other defects.
7. Reconcile a known item's lookups across vertical, horizontal and INDEX/MATCH methods and write the range/column assumptions of each.

What should you see?

Known/missing/blank/padded-key test matrix with exact-method reconciliation. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Design an independent lookup output with Rate, MatchCount and Decision for the mixed intake. A zero rate, unknown key and duplicate key must remain distinguishable.
2. Move a column only on a disposable mapping copy and inspect which lookup formula needs repair. Explain why the original column-index assumption failed.
3. Hand over a rejected-key list and source-mapping check, leaving baseline Rates unchanged.

Evidence to Hand Over

- Known/missing/blank/padded-key test matrix with exact-method reconciliation.
- Duplicate-key and column-move diagnosis on labelled copies.
- Independent lookup result that never silently turns unknown rates into zero.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A lookup returns a plausible but wrong rate	Inspect exact-match mode, key uniqueness, whitespace and the lookup column position.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Reliable lookups depend on clean keys, explicit match behaviour and deliberate error handling.

Check yourself: Does an exact-match lookup prove that a matched source value is populated?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
LookupRate
```

```
=VLOOKUP(D2,Rates!$A$2:$D$7,3,FALSE)
```

```
=INDEX(Rates!$C$2:$C$7,MATCH(D2,Rates!$A$2:$A$7,0))
```

```
=HLOOKUP(F4,$G$1:$L$2,2,FALSE)
```

```
=IFNA(VLOOKUP("IT99",Rates!$A$2:$D$7,3,FALSE),"Unmapped item")
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Lookup challenge: Known IT01 and IT06; unknown IT99; blank; space-padded IT03. Duplicate-test mapping adds IT03 with UnitCost 99 only on a disposable copy. Baseline IT03 remains 30. Keep the supplied six-key Rates source intact.
```

06

CHAPTER 6

Complete the Formula Clinic

In this chapter you will debug supplied defects, explain the correction and reconcile a checked sample.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 65 minutes: 15 explanation/demonstration, 22 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 50 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: Three repaired defects with independent baseline reconciliations.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Diagnose formula defects.
- Test zero and missing values.
- Reconcile source totals independently.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use a challenge copy of your advanced workbook. Keep the baseline original untouched.

6.1 Create Three Test Defects

1. On the challenge copy, remove FALSE from a copied VLOOKUP, shift one multiplication reference to the next row, and shorten one SUM range by a row.
2. Record the intended formula behaviour before fixing anything.
3. Inspect the formula bar, not just the visible result, for each defect.

What should you see?

You have three clearly identified test defects; a plausible number is not assumed correct.

6.2 Repair and Test

1. Restore exact matching and the correct same-row multiplication. Extend the total to every source row.
2. Test a temporary zero quantity and check that its amount becomes a valid zero. Restore the quantity.
3. Test an unknown item code in a separate practice cell and confirm it remains an explicit missing match.

What should you see?

The repair handles ordinary rows, valid zero and a missing key distinctly.

6.3 Reconcile and Explain

1. Sum line amounts from all 24 rows and reconcile to 3,158. Filter Delivered rows and reconcile their amount to 1,478.
2. Compare the conditional summary with a manually inspected subset.
3. Write one sentence per defect explaining the cause and why your test detects it.

What should you see?

The baseline is restored and both independent control totals agree.

6.4 Run an Independent Formula Fault Clinic

1. Create a defect register for wrong match mode, shifted row reference and omitted total row. Include expected behaviour, observed symptom, hypothesis and proof.
2. Preserve faulty and repaired formulas side by side on the clinic copy so a reviewer can see the specific change.
3. Use a small manually calculated subset to distinguish a logic error from a display-format difference. Do not repair by typing the desired result.
4. Add the duplicate-key and numeric-text cases from the card as separate tests. Decide whether each problem belongs in source cleaning, lookup validation or formula logic.
5. For every repair, test one normal input and one boundary/exception input. Record actual results and any unresolved issue.
6. Reconcile baseline all-orders and Delivered totals after restoration, plus a record-count control. Keep cancelled rows in the source.
7. Write a brief troubleshooting decision path: inspect source → define population → inspect references → test boundary → reconcile result.

What should you see?

Defect register with preserved faulty/repaired formulas and test evidence. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Ask a partner to introduce a defect on a separate clinic copy without telling you its cell; if alone, use the card and hide your defect note before diagnosis.
2. Locate it through controls, isolate the cause, repair it and run a regression check on a previously correct population.
3. Present the evidence trail in two minutes, including why your chosen control would catch the defect on a future refresh.

Evidence to Hand Over

- Defect register with preserved faulty/repaired formulas and test evidence.
- Normal/boundary checks for every repair and restored baseline controls.
- Independent diagnosis, regression check and an explainable troubleshooting path.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The total is corrected but the defect remains	Require a formula/source repair and a new input test rather than a hard-coded total.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

A useful formula check tests meaning and boundaries rather than merely the absence of an error message.

Check yourself: Which of your checks would catch a missing final row?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Extra fault choices, clinic copy only: change a numeric quantity to text “four”; insert a duplicate lookup key; replace a blank/missing-rate outcome with 0; shorten one delivered summary range by two rows. Diagnose the cause before restoring from Source Snapshot.
```

07

CHAPTER 7

Summarise with Pivot Tables

In this chapter you will group project costs and status; inspect filters, calculated fields and source coverage.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 75 minutes: 15 explanation/demonstration, 32 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 60 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A native pivot with checked filters and a calculated-field demonstration or recorded limitation.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Create a native pivot summary.
- Choose filters and aggregation deliberately.
- Check a calculated field against row calculations.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders A1:K25 after calculating Amount in J and ActiveAmount in K. A recovery route is to repeat Advanced Chapter 2 on the supplied workbook.

7.1 Build the Pivot

1. Select Orders A1:K25 and use Insert, Pivot table. Place it on a new tab named `Pivot Review`.
2. Add Project as rows and Status as columns. Add Amount as values, summarised by SUM.
3. Inspect totals and confirm the grand total is 3,158.

What should you see?

The pivot shows all projects and statuses. It has not discarded cancelled records without a stated filter.

7.2 Filter and Compare

1. Filter the pivot to Delivered only and check the total becomes 1,478.
2. Compare the result with the earlier SUMIFS or filtered source calculation.
3. Remove the pivot filter and verify the all-orders total returns.

What should you see?

Filtered and unfiltered results answer different documented questions.

7.3 Inspect a Calculated Field

1. Create a separate practice pivot using the same source. Group by Project.
2. Add a calculated field for average line amount, using `=SUM(Amount)/COUNTA(OrderID)` and **Summarise by**, **Custom** if supported by the target UI. Compare with a source calculation. Retain SUM of the row-level Amount as the authoritative total.
3. If that calculated-field expression is rejected, retain the working Amount source column and record the limitation for the trainer; do not replace it with an unrelated metric.

What should you see?

The calculated-field demonstration is reconciled to the existing row calculation, or explicitly recorded as unavailable.

7.4 Reconcile Pivot Semantics and New-Row Coverage

1. Create a second pivot counting OrderID by Project and Status. Compare its population count with the 24-row source and its category counts with COUNTIFS.
2. Double-click a populated pivot value where supported to inspect its contributing source rows. Reconcile the amount from those rows independently.
3. Use a safe calculated field such as `=SUM(Amount)/COUNTA(OrderID)` with **Summarise by**, **Custom**, where the target UI supports it. Label this as average line amount and compare with a source calculation.
4. Do not replace SUM of row Amount with SUM(Quantity)*SUM(UnitCost). Calculate both on a small two-row practice table and explain the different meanings.
5. On a separate pivot challenge copy append the supplied PO025 record. Inspect whether the fixed pivot source range includes it.
6. Extend the source range and inspect both the count and amount pivots. Compare them with a repaired formula summary.
7. Record pivot source range, filters, aggregation and calculated-field definition in an analysis note. If the calculated field is unavailable, retain source Amount and document that specific UI check as pending.

What should you see?

Amount and count pivots reconciled to formulas and contributing source rows. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Build a fresh pivot independently answering Delivered amount by Project, with a count of contributing order lines.
2. On the challenge copy include PO025 and show the source row, count, total and changed project contribution. Leave the baseline pivot unchanged.
3. Explain one aggregation trap and use contributing rows to prove the correct result.

Evidence to Hand Over

- Amount and count pivots reconciled to formulas and contributing source rows.
- Safe calculated-field comparison or documented unavailable route.
- Separate new-record pivot with repaired coverage and demonstrated aggregation semantics.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A pivot total is wrong even though the row appears	Inspect aggregation, filters, source range and calculated-field meaning separately.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Native pivots summarise a defined source. Their filters and formulas need the same checks as ordinary calculations.

Check yourself: Why must the pivot's source range include every record and calculated column?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Pivot Review
```

```
=SUM(Quantity*UnitCost)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
PO025, challenge copy only: PJ-A | Planning | IT01 | Quantity 2 | UnitCost 45 | Delivered |
week 4 | Additional practice line. Amount is quantity times cost. Extend both formulas and
pivot source range; do not alter the baseline workbook.
```

08

CHAPTER 8

Query Structured Criteria

In this chapter you will use DSUM, DCOUNT and DAVERAGE with a visible criteria range and reconcile against a manual subset.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 60 minutes: 15 explanation/demonstration, 17 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A visible criteria area and reconciled database function results.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Build a visible database criteria range.
- Use DSUM, DCOUNT and DAVERAGE.
- Reconcile criteria-based results.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use Orders A1:J25, including Amount. Keep criteria outside the source table.

8.1 Build the Criteria

1. On Analysis type Status in A10 and Delivered in A11. Match the Orders header exactly.
2. In C10 type `Delivered amount`. In D10 enter `=DSUM(Orders!A1:J25,"Amount",A10:A11)`.
3. Compare the result with the delivered control total.

What should you see?

DSUM returns 1,478 using the visible Delivered criterion.

8.2 Count and Average

1. In C11 type `Delivered numeric quantity records`; in D11 enter `=DCOUNT(Orders!A1:J25,"Quantity",A10:A11)`.
2. In C12 type `Average delivered unit cost`; in D12 enter `=DAVERAGE(Orders!A1:J25,"UnitCost",A10:A11)`.
3. Discuss why DCOUNT of the text OrderID field would not count order IDs like a text-aware count.

What should you see?

DCOUNT counts numeric Quantity entries in qualifying records. The average is of unit costs, not a quantity-weighted price.

8.3 Check Criteria Semantics

1. Filter the source to Delivered and manually inspect a small sample contributing to each result.
2. Add Project as a second criteria header beside Status and use PJ-A on the same criteria row to mean both conditions.
3. Compare with SUMIFS for the same project/status pair. Restore or clearly label the expanded criteria.

What should you see?

The database function and ordinary conditional formula agree when their criteria describe the same subset.

8.4 Prove AND, OR and Overlapping Criteria

1. Create three separate criteria blocks with exact source headers: Delivered AND PJ-A; Delivered OR Pending; and Delivered OR PJ-A. Label each natural-language population.
2. Use same-row criteria for the AND block and separate criteria rows for OR blocks. Keep blocks spatially separate from result cells.
3. Calculate DSUM, DCOUNT of numeric Quantity and DAVERAGE of UnitCost for each block. Record how blanks or text could affect the latter two metrics.
4. For the AND block reconcile amount with SUMIFS and list matching OrderIDs using a filter view.
5. For Delivered OR Pending reconcile with the sum of two disjoint status totals. For Delivered OR PJ-A inspect the overlap so a record satisfying both conditions is not counted twice.
6. Compare an unweighted mean UnitCost with a quantity-weighted cost measure on a small explicit test. State which question each answers.
7. Deliberately misspell one criteria header on a disposable block. Record the observed error or result and repair it using the exact source heading.

What should you see?

Three labelled criteria blocks with source-ID reconciliation. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Write a new analysis question requiring two conditions and design its database criteria without copying a finished block.
2. Create a second version that uses OR instead of AND, predict the population difference and verify it through contributing IDs.
3. Explain the numeric field counted by DCOUNT, the average's denominator and one overlap risk.

Evidence to Hand Over

- Three labelled criteria blocks with source-ID reconciliation.
- AND/OR and overlap proof, plus weighted/unweighted meaning comparison.
- Independent question and corrected criteria-header test.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
An OR total double-counts overlapping rows	Compare the contributing ID sets and count each qualifying source record once.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

Database functions are useful when their header matching, criteria logic and field meaning are explicit.

Check yourself: Why is an average unit cost different from a quantity-weighted average?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
Delivered amount
```

```
=DSUM(Orders!A1:J25,"Amount",A10:A11)
```

```
Delivered numeric quantity records
```

```
=DCOUNT(Orders!A1:J25,"Quantity",A10:A11)
```

```
Average delivered unit cost
```

```
=DAVERAGE(Orders!A1:J25,"UnitCost",A10:A11)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Criteria cases: Status=Delivered AND Project=PJ-A; Status=Delivered OR Status=Pending;
Status=Delivered OR Project=PJ-A. The last case overlaps. DCOUNT uses numeric Quantity;
OrderID is text and must not be treated as a numeric record count.
```

09

CHAPTER 9

Build a Dashboard

In this chapter you will choose charts, sparklines and filter views; distinguish source-linked charts from generated analysis snapshots.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 70 minutes: 15 explanation/demonstration, 27 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 55 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A compact source-linked dashboard with a tested update path.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Choose source-linked visuals.
- Use a sparkline for a defined time sequence.
- Explain how a dashboard refreshes.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use the checked advanced workbook with all-orders and delivered controls reconciled.

9.1 Build a Compact View

1. Create a Dashboard tab. Link a small project summary to the checked analysis or create a fresh pivot from the same source.
2. Add a chart comparing Amount by Project and label whether cancelled orders are included.
3. Add a separate status count view with clear labels. Keep detailed source records in Orders.

What should you see?

The dashboard identifies the question and population behind each number.

9.2 Show the Weekly Sequence

1. On Analysis create week labels 1, 2, 3 and 4 and calculate Amount totals for each with SUMIFS.
2. On Dashboard create a labelled sparkline using the four weekly result cells, for example `=SPARKLINE(Analysis!B16:E16)` after placing the totals there.
3. Check that the sequence is chronological and its totals reconcile to all-orders Amount.

What should you see?

The sparkline describes the four supplied practice weeks, not an invented real calendar period.

9.3 Test a Refresh

1. Temporarily change one source quantity. Check the line amount, summary, chart and sparkline.
2. Inspect whether a pivot has updated and whether its source still includes all rows.
3. Restore the source quantity. Document any manual refresh step and keep filters visible or clearly named.

What should you see?

The dashboard returns to the reconciled baseline after restoration.

9.4 Design a Dashboard With a Visible Refresh Contract

1. Write a dashboard brief specifying audience, population, units, three useful questions and the source range. Do not mix all-orders and active totals without labels.
2. Add KPI cells for record count, all-orders Amount, ActiveAmount and Delivered Amount. Reconcile each to the corresponding checked analysis.
3. Show project amount comparison, status count and weekly sequence with explicit filter scope. Keep week order numeric and label it as practice week.
4. Create a visible **Checks** block comparing KPI totals with source controls and indicating any mismatch. It must recalculate from cells rather than contain typed “OK” text.
5. On a scenario copy change PO001 Quantity from its baseline to 4. Predict the amount delta and check the source line, summary, chart and weekly sequence.
6. Restore the quantity and prove all affected values return. Record which components follow source-cell changes and which depend on range extension for new rows.
7. Prepare a refresh contract listing source bounds, pivot bounds, formula bounds, chart ranges, filter scope and snapshot/export steps.

What should you see?

Defined dashboard with distinct populations, units and recalculating controls. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Build an independent dashboard view answering the three audience questions on the card, with at least one visible reconciliation check.
2. Append PO025 only on a challenge copy and repair source-range coverage across KPIs, pivots, charts and the weekly sequence.
3. Ask a reviewer to identify the population and a stale component using the labels/checks, then revise any misleading element.

Evidence to Hand Over

- Defined dashboard with distinct populations, units and recalculating controls.
- Source-change/restoration trace across line, summary and visuals.
- Independent new-row coverage test and a usable refresh contract.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A dashboard looks current but omits the new line	Trace range coverage through every KPI and visual rather than checking only one total.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

A dashboard is useful when its totals, populations and update path are understandable.

Check yourself: Which dashboard objects depend on which source calculations?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 1: Typed Values

```
=SPARKLINE(Analysis!B16:E16)
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Audience questions: Which project has the largest Delivered amount? How much Amount remains
Pending? Which practice week has the largest all-orders Amount? PO025 extension: PJ-A,
Planning, IT01, Quantity 2, UnitCost 45, Delivered, week 4.
```

10

CHAPTER 10

Use Gemini as an Analysis Partner

In this chapter you will generate and inspect formula proposals and chart summaries; test exceptions before accepting.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 55 minutes: 10 explanation/demonstration, 25 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 45 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A checked AI formula proposal and evidence-based findings.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Ask AI for a bounded formula proposal.
- Verify a generated finding.
- Distinguish a source-linked result from a generated snapshot.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use the native checked workbook and usable Gemini in Sheets access. If unavailable, critique the supplied examples manually and record the AI objective as incomplete.

10.1 Request a Formula Proposal

1. Open Ask Gemini. Paste the Part 2 prompt describing the actual source columns and range.
2. Read the proposed formula before applying it. Check its row bounds, exact status spelling and use of Amount.
3. Test the formula in a spare labelled cell and compare with the known Delivered total of 1,478.

What should you see?

The proposal is accepted only after it matches the actual schema and control total.

10.2 Check an Interpretation

1. Ask for two findings using the checked project/status summary, without web research.
2. For each finding, identify the cells that support it. Reject unsupported causal claims, forecasts or benefit percentages.
3. Write a final finding in your own words with its population and units.

What should you see?

Each retained finding is traceable to values rather than AI confidence.

10.3 Inspect Generated Visuals

1. If asking Gemini to generate a chart, inspect where its data was written and what range the chart references.
2. Temporarily change a source quantity and observe whether that generated analysis follows the original source.
3. Restore the quantity. Label a static/generated analysis snapshot appropriately or use your already checked source-linked chart.

What should you see?

The final dashboard does not imply automatic refresh where it has not been demonstrated.

10.4 Evaluate AI Proposals Against Known Test Cases

1. Prepare a test matrix before prompting: Delivered total, Pending >=200, unknown ItemCode and a new record outside the original bounds.
2. Ask Gemini for a formula and explanation using the actual headers and range. If unavailable, use the deliberately flawed proposals on the card for the same audit.
3. Inspect every proposed formula's population, field, match mode, bounds and copied-reference behaviour before using it.
4. Run the proposal in labelled spare cells against known controls and a manually verified subset. Record pass/fail and the cause of each failure.
5. Request a correction for one identified defect, or repair it manually. Keep the original proposal alongside the checked replacement so improvement is demonstrable.
6. For any generated chart inspect its data source and test a source-cell change. Record whether it follows the maintained data, separate generated cells or a static snapshot.
7. Write a final two-finding commentary grounded in checked values, rejecting causal claims, forecasts and unsupported ROI.

What should you see?

Prewritten test matrix and observed proposal results. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Design an independent prompt that includes the population, fields, intended output and tests, then run or manually review its result.
2. Challenge it with IT99 and PO025 in separate test contexts. Keep missing mapping distinct from zero and inspect new-row coverage.
3. Produce a short audit comparing original proposal, correction, evidence and remaining limitation.

Evidence to Hand Over

- Prewritten test matrix and observed proposal results.
- Original/corrected formula trace with independent verification.
- Generated-chart source check or stated unavailable route, and supported final findings.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The AI explanation sounds right while the formula differs	Audit the actual formula and contributing rows, not its prose description.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

AI can propose analysis; source tests establish whether it is suitable.

Check yourself: What evidence would establish that a generated chart follows the maintained source?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

Part 2: Propose a Delivered Total

Session: new chat | **Grounding:** explicit Orders schema

```
Goal: Propose a formula for delivered order amount.
Context: Fictional Google Sheets analysis.
Source: Orders rows 2 to 25; Status is G and Amount is J. Header row is 1.
Expectations: Reply with a formula and explanation only. Do not edit cells. Include only
Status equal to Delivered. Use bounded ranges. I will test the formula against the source.
```

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Flawed formula proposals to audit if AI is unavailable: `=SUM(Orders!J2:J24)` described as all
24 rows; `=IFERROR(VLOOKUP("IT99",Rates!A2:D7,3,FALSE),0)` described as a confirmed zero rate;
`=SUMIFS(Orders!J2:J25,Orders!G2:G25,"Delivered")` described as active amount excluding only
Cancelled.
```

11

CHAPTER 11

Complete the Analysis Capstone

In this chapter you will answer a new project question, substantiate the finding and deliver a checked dashboard.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 80 minutes: 15 explanation/demonstration, 37 guided build and casework, 20 independent challenge, 8 verification and repair. The active portion is 65 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A source-traceable capstone analysis and checked presentation.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Answer a new analysis question.
- Reconcile a result independently.
- Communicate assumptions and limitations.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Create a challenge copy. Use the same 24-order source and six-item Rates table; do not import real purchasing data.

11.1 Define and Calculate

1. Answer: which project has the largest Delivered amount, and which Pending records have Amount at least 200?
2. Write the population and criteria before choosing formulas or a pivot.
3. Calculate the result and retain a visible trace to qualifying order IDs.

What should you see?

The ranking and exception list are based on explicit delivered/pending definitions.

11.2 Challenge the Result

1. Reconcile the delivered total to 1,478 and all-orders total to 3,158.
2. If no baseline pending row qualifies, record zero matches explicitly. Inspect a non-qualifying baseline row and use a separate synthetic row to test the positive boundary without altering the baseline.
3. Test a duplicate or missing lookup key on a disposable copy and explain its effect without changing the baseline workbook.

What should you see?

Your result survives an independent check and identifies meaningful data assumptions.

11.3 Present the Analysis

1. Show one clear chart or table with project names and currency units.
2. State the source population, result and one limitation. Do not equate order amount with profit or budget approval.
3. Explain which sources and calculations must update for a future record.

What should you see?

A reviewer can trace the finding to source records and understand its limits.

11.4 Complete a Decision Analysis With Independent Evidence

1. Write a one-page analysis specification for the card's three questions, naming population, metric, threshold and exclusions.
2. Build answers with formulas and a pivot or filtered-source reconciliation. Retain the contributing OrderIDs for the exception list.
3. Compare project Delivered amounts and identify the largest using checked numbers. Do not infer profitability or project performance from order amount alone.
4. For Pending Amount >=200, first report the actual baseline match count, including zero if none qualify. Inspect a non-qualifying baseline row and use a separate synthetic positive/boundary test. Keep synthetic tests outside the baseline analysis.
5. Run a separate PO025 scenario and describe its effect on ranking, delivered totals and week 4. Do not overwrite the baseline result.
6. Prepare an executive table/chart and an audit appendix showing source range, assumptions, controls and unresolved data issues.
7. Ask a reviewer to challenge one answer and prove it from the source rows and a second calculation method.

What should you see?

Three-question specification and independently reconciled results. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Independently compare all-orders and active populations and explain how excluding Cancelled changes the question and results.
2. Stress-test an unknown rate key and a duplicate lookup key in a scenario copy. Demonstrate the warning/disposition rather than forcing a numeric answer.
3. Deliver a three-minute recommendation limited to what the data supports, plus one next question requiring additional evidence.

Evidence to Hand Over

- Three-question specification and independently reconciled results.
- Separate added-row and key-quality scenarios with audit evidence.
- Executive output, reviewer challenge response and a bounded recommendation.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
A recommendation claims savings or profitability	Return to dataset meaning and request missing cost/outcome evidence before making that claim.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

The capstone combines reliable data, explicit calculations and a defensible explanation.

Check yourself: Would the same result follow if Cancelled records were included?

Zero-Match Check

The supplied baseline has no Pending record with Amount at least 200. Record that result explicitly. Use a separately labelled synthetic case to prove the positive branch; do not edit baseline records just to obtain a highlight.

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Capstone questions: Largest Delivered amount by project; Pending records with Amount ⩾200 and
their IDs; highest all-orders Amount by practice week. Scenario PO025: PJ-A, Planning, IT01,
Quantity 2, UnitCost 45, Delivered, week 4. No actual budget or profit data is supplied.
```

12

CHAPTER 12

Review and Handover

In this chapter you will explain assumptions, refresh steps and known data limitations.

> **Prompts and values:** use the copy-paste page for the exact values used below.

Target allocation: 50 minutes: 10 explanation/demonstration, 20 guided build and casework, 15 independent challenge, 5 verification and repair. The active portion is 40 minutes by design. These are unmeasured targets awaiting a learner pilot.

Your result: A restored baseline and an update-tested workbook handover.

> **Status:** expanded review draft, revision 0.4. Complete the casework and independent challenge as well as the guided steps. The expanded route, screenshots and timing have not yet been validated in the learner application.

What You Will Learn

- Explain the workbook update path.
- Identify future maintenance work.
- Leave a checked final state.

Before You Begin

Read the case input and use a separate named casework copy for destructive tests or altered datasets. Keep the baseline unchanged for later chapters. Record source facts, proposed practice and unresolved questions separately.

Use your baseline workbook and separate capstone copy.

12.1 Review the Baseline

1. Clear temporary test entries and restore all source values changed during exercises.
2. Confirm the 24-record count and control totals.
3. Inspect filter states and remove or name any view that could mislead the next reader.

What should you see?

The baseline is consistent and temporary tests no longer affect it.

12.2 Write the Handover

1. Record where the source lives, what one row represents and which columns users may edit.

2. Describe formula, validation, pivot and chart ranges that must include a new row.
3. Document unmapped keys and other unresolved data issues rather than converting them to zero.

What should you see?

A future editor can update the analysis without silently omitting records.

12.3 Demonstrate One Update

1. On the capstone copy, add one new fictional order line and walk through the documented update steps.
2. Check the resulting calculations and visuals, then clearly label that copy as an extension test.
3. Keep the original baseline for future class recovery.

What should you see?

The handover has been tested on an actual extension in a separate copy.

12.4 Make the Analysis Operable by Another Analyst

1. Create a **Runbook** with source location, row grain, accepted inputs, editable fields, formula columns, exception handling and release controls.
2. Create a range inventory for validation, conditional formatting, SUMIFS, lookups, database functions, pivots, chart sources and weekly summaries used in your workbook.
3. On a separately named extension copy add PO025 and follow only the runbook to extend the analysis. Record every instruction that needed interpretation.
4. Reconcile the 25-record scenario by count, all-orders total, Delivered total and project/week contribution, then compare with the preserved 24-record baseline.
5. Test the unknown-key route without overwriting the supplied Rates mapping. Record the issue and leave a visible unresolved disposition.
6. Give a partner the runbook and ask them to locate a source, run a check and explain a failed key. If alone, use a fresh read-through with previous working tabs closed.
7. Repair one handover gap and issue a manifest identifying the maintained baseline, scenario test and current fixed summary.

What should you see?

Complete range inventory and a runbook tested on PO025. The result must be supported by the recorded checks; an attractive output or an AI assurance is insufficient.

Independent Challenge

Close the earlier step lists for the first attempt. Use the case input and your own reasoning. If you need help, record the hint and repeat the affected check independently.

1. Use the different PO026 line on the card in another disposable copy and update the analysis without returning to the lesson steps.

2. Verify how Pending amount and exception flags change, including any new-row coverage failure. Record and repair the cause.
3. Hand over the runbook, controls, open issues and manifest. Another analyst must be able to reproduce both a successful update and an unresolved-key outcome.

Evidence to Hand Over

- Complete range inventory and a runbook tested on PO025.
- Independent PO026 update with reconciled controls and exception coverage.
- Recorded handover repair, open-issue route and clear scenario manifest.

Record the defect or uncertainty you investigated, the evidence used, the repair or reasoned no-change decision, and the final state. Restore temporary changes or keep them in a clearly named scenario copy.

Troubleshooting

Symptom	What to check
The next analyst must guess a range extension	Record exact current bounds, the new-row action and the expected checks in the runbook.
An input or earlier result is missing	Reopen the input named in Before You Begin; do not substitute private company data
A menu or feature is unavailable	Confirm the account and the shared access notes; record the missing step and ask the trainer for its fallback
The result differs from the checkpoint	Compare the exact source range, wording and conditions before repeating the action

Lesson Summary

You can maintain and hand over a checked analysis rather than a collection of unexplained formulas.

Check yourself: Which assumption would you check first if a future total unexpectedly drops?

Prompts and values

Copy the values below when following the chapter. Review pasted text before submitting it.

This chapter uses the supplied files and visible controls. No additional prompt or typed code is required.

Depth Case Input

Fictional training material. This is exercise content to analyse, not authorisation to send, publish, grant access or change a real project decision.

```
Independent PO026, separate baseline-derived copy: PJ-B | Procurement | IT03 | Quantity 10 | UnitCost 30 | Pending | week 4 | Handover exception test. Do not combine it with PO025 unless you explicitly define and reconcile that additional scenario.
```

Google Sheets Advanced

A practical 2-day course with step-by-step draft labs, copy-ready prompts, fictional inputs and expected-result checks. Use the supplied practice pack. Full lab execution, screenshots and classroom timing validation remain required before participant release.

Asraf Jaafar Sidik · Microsoft Certified Trainer ·